A low speed of vibration equates to a lower-frequency sound, while faster vibrations make higher frequency sounds. So when humans talk of sound and frequency, they normally refer to the range between 20Hz and 20,000 Hz that the human ear can hear. When a sound's frequency is on the lower end of our hearing scale (down near 20hz), we hear deep/low notes/sounds which we commonly call Bass. When the frequency is up near 20,000 Hz, we hear high-pitched sounds which we commonly call Treble. We call the middle ground frequency midrange.

How do loudspeakers work, then?

So logically, the only way a speaker would be able to reproduce sound would be to reproduce its original mechanism of production i.e. by vibrating the surrounding air medium. Since the information regarding the sound is passed on electronically, the speaker will convert electrical energy to acoustical energy. The electronic signal would dictate the frequency of the note, and the speaker would faithfully reproduce that in a mechanical form by moving back and forth, thus increasing and decreasing the air pressure in front of it and creating sound waves.

The cone, or the diaphragm, is the main moving mass of the speaker. Vibrating this cone, which is almost like a drum skin, is what produces and amplifies the sound in a loudspeaker. So it makes sense that if the cone is larger, the speaker will have more mass and surface area. The more surface area a speaker has, the more air it can move. The more air it can move, the louder the speaker can get.

The ideal material for the diaphragm would need to be rigid, so as to prevent uncontrolled cone motions; have low mass to minimize the amount of initial energy needed to get it vibrating initially; and be well damped, which essentially means that the time period needed to shift the vibration in accordance with the frequency should be short, with little or no audible ringing due to its resonance frequency. Sound engineers haven't been able to find one single material that satisfies these criteria, so they have turned to using composite materials, such as glass, Kevlar, carbon fibre etc.

The outer rim of the cone is fixed to an enclosure of sorts, which is generally the loudspeaker's circular rim through a spring or some flexible suspension called a spider, to allow it to vibrate in place. The inner end of the cone is affixed to an iron coil, or an electromagnet, which is again placed in front of a permanent magnet, which is fixed in place. The electromagnet, on the other hand, is free to move. As electric pulses pass through the coil of